Uddhav Panchavati

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Education

University of California, San Diego (UCSD) — Sept 2024 – Jun 2028
B.S. Cognitive Science (Machine Learning & Neural Computation), Minor in Data Science — *La Jolla, CA*

Work Experience

Research Intern — Jul 2024 – Present
UCLA BAIR Lab — *Los Angeles, CA*

- Placed 20th out of 8000 submissions at the NeurIPS EEG Foundation Model Challenge. (2025)
- Designed and evaluated an EEG foundation model with a spatiotemporal graph-based sequence architecture incorporating novel graph-walk algorithms; leveraged AWS for large-scale data management and pretraining.(2024)
- Developed dataset and training pipelines across 25 EEG datasets; implemented and benchmarked custom and state-of-the-art models to evaluate performance and generalizability.
- Evaluated and optimized thyroid cytology stain normalization models to improve histological consistency and reliability across datasets.

Machine Learning Research Intern — Jun 2023 – Nov 2023
Conifer Point — *Remote*

- Designed protein candidates for Alzheimer's treatment using Boltzmann Maps and Schrödinger; optimized binding efficiency and structural stability.
- Developed ML-based energy estimation shortcuts to bypass traditional protein energy calculations, reducing drug design runtime and improving model scalability.

Extracurricular Activities

Co-Founder & Full Stack Developer — Jan 2025 – Present
TritonPlanner

- Scraped and cleaned 7,000+ UCSD course listings using Selenium automation; built Pinecone vector databases for course research, querying, and hierarchical prerequisite organization.
- Developing LangChain multi-agent pipelines for parsing transcripts and generating curated schedules based on major and college requirements.
- Helped build a Node.js backend and interactive React interface supporting a drag-and-drop 4-year planner, course search/assistant, and graduation requirements sidebar.

Project Lead — Nov 2024 – Present
Triton Neurotech (De Sa Lab)

- Developed an fMRI-to-image reconstruction method by contrastively aligning fMRI embeddings with a pretrained VQGAN image encoder, enabling reconstruction via the VQGAN decoder; presented at the California Neurotechnology Conference poster session.
- Fostered collaboration with De Sa Lab to research multimodal neural image decoding.

Epilepsy Uncertainty Framework — May 2024 – Jun 2024
Elkins High School

- Developed a deep learning model to predict seizure onset as probability distributions.
- Achieved 86.9% sensitivity and reduced false positives to 0.063/hour on the CHB-MIT EEG dataset, outperforming conventional CNN-based baselines.

Winner — AI Hackathon: Transforming Rare Disease Care — Apr 2025
UC Berkeley Blum Center & Haas School of Business

- Led development of an AI agent that converts physician notes and EMRs into reimbursement-ready documentation with CPT codes by auto-generating evidence-backed clinical reports.
- Engineered a literature-mining module that extracts disease-specific data and synthesizes findings into concise supporting documents for rare disease care.

Skills

Languages: Python, Java, C++
ML/DS: PyTorch, TensorFlow, NumPy, Pandas, scikit-learn, MNE
Web/Infra: Node.js, React, SQL, AWS, Pinecone, LangChain
Certifications: AutoCAD Certified, OSHA Certified